PATTERN FORMATION IN NATURE

INTRODUCTION

Have you wondered how patterns are formed on animal skin? Do you wonder what is the purpose of patterns? Why do complex patterns like fractals exist?

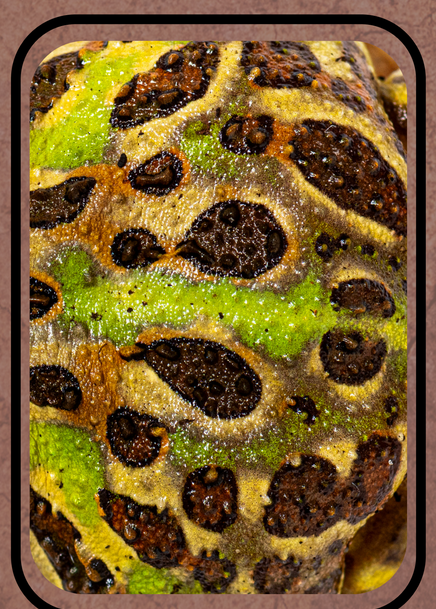
FORMATION OF PATTERNS IN ANIMALS

According to Turing's pattern theory; patterns are formed due to reaction-diffusion reactions, where;

- Slow diffusing activators that produces pigment is suppressed by relatively fast diffusing inhibitors → forming visible pigment → forms patterns

REACTION-DIFFUSION IN ANIMALS

Zebra stripes are an example of a Turing reaction-diffusion system. The uniform skin cells undergo embryogenesis, where the relatively strong inhibitors prevent spots from forming next to each other, causing lines to form.



NATURAL SELECTION

Animals who gain beneficial patterns (such as camouflage) through evolution are more likely to survive and reproduce, passing down these patterns on to the next generation.
Animals without these patterns fail to adapt and are more likely to die out.

FRACTALS IN... NATURE

WHAT ARE FRACTALS?

Fractals are infinitely complex patterns that are self-similar across different scales.

REFLECTIONS IN NATURE

Fractrals are the most efficient way for systems to grow, distribute resources and fill space.

For example, roots often branch out to maximise their exposed surface area to volume ratio.

CONCLUSION

We can summarise pattern forming with a simple analogy. If a cheetah skin is a forest, activators are fires that spread slowly, and inhibitors are firefighters that put out the fires. The black ashy spots left after the fires are the patterns that form on animals.